

Phase Transition Geometry on the Generator-Set Moduli Space

Global Stratification, Local Commutativity Geometry, and Accessibility Walls

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This paper is Part VI of the RIME program. Papers I–V study static structures at fixed generator sets. Paper VI opens the deformation line: what happens when the generator set itself varies? The paper studies the deformation geometry of the generator-set moduli space and the wall structure of accessibility phase transitions.

Abstract

Problem. Why do sectors and accessibility observables move? Paper IV treats a fixed joint arrangement P and varies only the projection. Under generator-set perturbation the arrangement itself becomes $P(w)$, and it is defined only where $Q_T(w)$ and $H_T(w)$ form a commuting normal pair. The problem is to separate the spectral walls of this moving arrangement from the accessibility walls of the induced sectorized system.

Approach. We introduce the generator-set moduli space $\mathcal{M} = [0, 1]^m$ and distinguish two observables. The collision quotient is defined on normal spectral charts $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$, where $\Sigma_{\text{comm}} = \{w : [Q_T(w), H_T(w)] = 0\}$ and a QT/HT joint arrangement exists. The A-spectrum of the global average $A(w)$ is defined on all of $\mathcal{M}$. Locally, we linearize the commutator map at the canonical point $w = \mathbf{1}$ and compute the tangent model of Σ_{comm} . On normal charts of this base we define the moving sectorized system, its block fields, the accessibility jet, and the discrete shadows $R_1(w), R_2(w), D(w)$.

Results. The global scans separate the collision quotient from the A-spectrum and organize internal events as SPLIT, MERGE, and FIELD. At the canonical Rubik point, the linearized commutator map has rank 11 and a 7-dimensional kernel, decomposing as 1(HT) + 6(QT). In the tested local model, rank stability and nonlinear-kernel searches support a computational 7-dimensional tangent-local scaffold for Σ_{comm} near $w = \mathbf{1}$. On normal charts of this base, fragmentation of joint sectors produces jumps in the observable shadows R_1, R_2, D .

Implications. Spectral walls and accessibility walls are different discriminant structures. Σ_L and Σ_{field} describe movement and arithmetic of the joint arrangement; Σ_{access} records rank/support failures of the accessibility jet. This turns generator-set variation into the moduli-space form of the gap between Lie-generated and composition-generated accessibility.

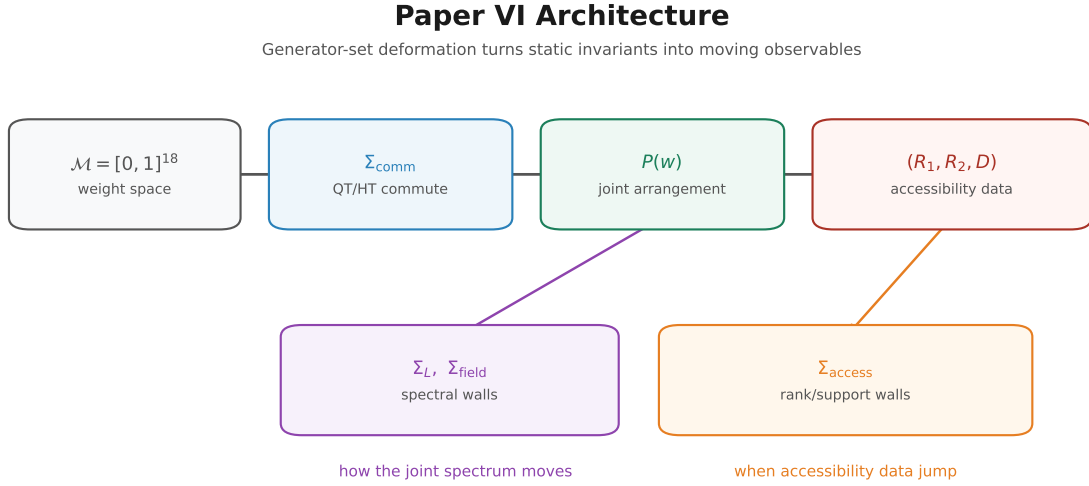
1 Structure of the Paper

Why do sectors move? Because a generator-set perturbation changes the averaged operators from which the joint arrangement is computed. On normal spectral charts $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$, that moving arrangement produces moving sector projectors; the observables R_1, R_2, D are discrete shadows of their block and commutator fields. The paper studies where these shadows remain stable and where they jump.

The paper has five structural layers. First, the commutativity locus Σ_{comm} is treated through local smooth models on which the QT/HT joint arrangement is defined. Second, the accessibility

fields over this local base are introduced: generator blocks, projected commutator blocks, and Hall projection blocks. Third, the discrete observables R_1, R_2, D are recovered as rank/support shadows of these fields, and accessibility walls are defined as failures of local continuation of this shadow. Fourth, the spectral walls $\Sigma_L, \Sigma_{\text{field}}$ and the accessibility wall Σ_{access} are separated as different discriminant structures inside $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$. Finally, the resulting wall picture is interpreted as the moduli-space form of the gap between Lie-generated and composition-generated accessibility.

The central organizing point is that, within the tested tangent-local model, the spectral arrangement moves smoothly, while the associated accessibility data are discrete shadows of smooth matrix fields and can jump across rank/support walls.



Spectral walls are not accessibility walls by definition; they live on different shadows of the same moving arrangement.

Figure 1. Paper VI architecture. The generator-set moduli space contains the commutativity locus, where the QT/HT joint arrangement and its collision quotient are defined. Spectral walls and accessibility walls are different observable shadows of the moving sectorized system.

2 Notation

Symbol	Meaning
$\mathcal{M}$	generator-set moduli space $[0, 1]^m$; $m = 18$ for Rubik
$w = (w_1, \dots, w_m)$	generator weight vector
$Q_T(w)$	weighted quarter-turn average
$H_T(w)$	weighted half-turn average
$A(w)$	weighted full generator average
$P(w)$	QT/HT joint spectral arrangement when defined
$L_\alpha(q, h)$	affine projection $\alpha q + (1 - \alpha)h$
$C_{\text{comm}}(w)$	commutator map $[Q_T(w), H_T(w)]$

Symbol	Meaning
J_k	Jacobian matrix $\partial C_{\text{comm}}/\partial w_k _1$
I_{comm}	ideal generated by entries of $C_{\text{comm}}(w)$
Σ_{comm}	commutativity locus $\{w : [Q_T(w), H_T(w)] = 0\}$
Σ_{spec}	normal commutative spectral domain where QT/HT joint sectors and orthogonal projectors are used
Σ_L	layer-count wall inside Σ_{spec}
Σ_{field}	field-extension wall inside Σ_{spec}
SPLIT, MERGE, FIELD	internal bifurcation events on normal spectral charts
X_g	skew-Hermitian accessibility generator, usually $X_g = \log \rho(g)$ as in Paper V
$\mathcal{S}(w)$	sectorized operator system $(V, \{Q_i(w)\}, \{X_g\}_{g \in S})$
$Q_i(w)$	orthogonal joint-sector projector on Σ_{spec}
$B_{ij}^g(w)$	block map $Q_i(w)X_gQ_j(w)$
$\mathcal{J}_{\text{acc}}(w)$	accessibility jet: first-order data of block, commutator, and Hall projection fields
Σ_{access}	accessibility wall: rank/support discriminant of the accessibility jet
$R_1(w)$	generator-labelled directed support graph computed from block-field support
$R_2(w)$	repair graph computed from projected commutator survival
$D(w)$	minimal accessibility depth data

3 Theorem-Layer Overview

Spectral walls describe how the joint arrangement moves. Accessibility walls describe when the moved arrangement changes the gap between Lie-generated and composition-generated accessibility.

This distinction is structural. The walls Σ_L and Σ_{field} are spectral stratification walls: they record layer jumps and field changes of the QT/HT joint arrangement. They are not, by definition, accessibility walls. Accessibility walls are detected by changes in the accessibility jet and in its discrete shadows $R_1(w)$, $R_2(w)$, and $D(w)$ on normal spectral charts of the commutative locus.

The theorem-layer hierarchy is

$$\Sigma_{\text{comm}} \longrightarrow \mathcal{J}_{\text{acc}} \longrightarrow \Sigma_{R_1} \longrightarrow \Sigma_{R_2} \longrightarrow \Sigma_D.$$

The continuous object is the accessibility jet

$$\mathcal{J}_{\text{acc}} = (J_{\text{block}}, J_{\text{comm}}, J_{\text{depth}}).$$

The observables R_1, R_2, D are discrete projections of this jet, not continuous invariants themselves.

Part I – Global Stratification

4 From Fixed Collision Geometry to Moduli

Paper IV studies a fixed finite arrangement [5]:

$$P = \{(q_i, h_i)\} \subset \mathbb{R}^2.$$

For a fixed commuting pair (Q, H) , the interpolation

$$A(\alpha) = \alpha Q + (1 - \alpha)H$$

changes layers by changing the projection

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h.$$

Thus Paper IV is a fixed-arrangement theory:

$$P \text{ fixed, } \quad L_\alpha \text{ varies.}$$

In the Rubik laboratory, QT/HT are the canonical commuting averaged operators. The theory is formulated for arbitrary commuting averaged operators; QT/HT are the canonical Rubik instance used to compute and test the deformation picture.

Paper VI studies the next problem:

$$P = P(w).$$

The arrangement itself moves because the generator-set weights change the averaging operators from which the arrangement is computed.

5 Generator-Set Moduli Space

Definition 5.1 (Generator-Set Moduli Space).

Let $\rho : G \rightarrow U(V)$ be a fixed finite-dimensional unitary representation, and let

$$S = \{g_1, \dots, g_m\}$$

be a fixed ordered generator list. The generator-set moduli space is

$$\mathcal{M} = [0, 1]^m.$$

A point $w = (w_1, \dots, w_m)$ assigns a continuous inclusion weight to each generator. The vertices $\{0, 1\}^m$ correspond to ordinary generator subsets. The finite-group representation setting follows the standard semisimple background [23]; for the Rubik group instance we use the usual group-theoretic presentation and generator conventions [17].

For the Rubik system, $m = 18$: twelve quarter-turn generators and six half-turn generators.

Definition 5.2 (Weighted QT/HT Averages).

Let QT and HT denote the quarter-turn and half-turn subsets of S . Define

$$Q_T(w) = \frac{\sum_{g_i \in \text{QT}} w_i \rho(g_i)}{\sum_{g_i \in \text{QT}} w_i}, \quad H_T(w) = \frac{\sum_{g_i \in \text{HT}} w_i \rho(g_i)}{\sum_{g_i \in \text{HT}} w_i},$$

when the denominators are nonzero. If all weights in one class vanish, the corresponding average is treated as a degenerate boundary operator.

For arbitrary independent weights, these averages need not be Hermitian or normal: inverse-pair symmetry can be broken. The commutator equation alone therefore does not justify orthogonal joint-sector projectors. Whenever this paper uses QT/HT joint spectral sectors and orthogonal projectors, it is working on a normal chart of the commutative locus. In inverse-symmetric submoduli this normality is automatic; outside such charts, generalized nonnormal joint decompositions are not used here.

Thus the joint-spectral and accessibility-wall statements below are read on inverse-symmetric generator-weight charts, under which $Q_T(w)$ and $H_T(w)$ remain Hermitian.

Definition 5.3 (Collision Quotient on the Commutative Locus).

The QT/HT joint spectral arrangement at w is used in this paper only on the normal commutative spectral domain

$$\Sigma_{\text{spec}} = \{w : [Q_T(w), H_T(w)] = 0, Q_T(w), H_T(w) \text{ normal}\}.$$

On that domain, the commuting normal pair is simultaneously unitarily diagonalizable, and the joint spectrum is a finite weighted point set

$$P(w) = \{(q_i(w), h_i(w), d_i(w))\}.$$

Equivalently, the joint sectors are orthogonal primitive eigenspace sectors of the finite-dimensional commutative spectral algebra generated by $Q_T(w)$ and $H_T(w)$ on this normal chart. This is the same finite semisimple algebraic setting that underlies Bose–Mesner algebras and association schemes, used here as comparison classes rather than as an asserted identification [2, 11, 12].

For fixed α , the collision quotient is induced by

$$L_\alpha(q, h) = \alpha q + (1 - \alpha)h.$$

The layer count $N_L(w)$ is the number of distinct values of $L_\alpha(P(w))$.

Definition 5.4 (Global A-Spectrum).

The global averaged operator is

$$A(w) = \frac{\sum_i w_i \rho(g_i)}{\sum_i w_i},$$

for nonzero total weight. Its spectrum is defined on all of $\mathcal{M}$ away from the zero-weight vertex. This observable does not require QT/HT commutativity.

6 The Primary Commutativity Locus

The primary object of Paper VI is the commutativity locus. It is the algebraic wall controlling where QT/HT commutation holds. The QT/HT joint spectral arrangement and collision quotient are then used on its normal spectral charts $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$.

Definition 6.1 (Commutativity Locus).

The commutativity locus is

$$\Sigma_{\text{comm}} = \{w \in \mathcal{M} : [Q_T(w), H_T(w)] = 0\}.$$

For collision-quotient geometry, Paper VI further restricts to normal charts inside this locus. Outside Σ_{comm} , the QT/HT joint spectral arrangement is not defined in this framework; inside Σ_{comm} but outside normal charts, this paper does not invoke orthogonal joint-sector projectors.

On any chart where the total QT and HT weights are nonzero, the condition is equivalent to the vanishing of the polynomial commutator map

$$C_{\text{comm}}(w) = \sum_{i \in \text{QT}} \sum_{j \in \text{HT}} w_i w_j [\rho(g_i), \rho(g_j)].$$

The algebraic study of Σ_{comm} begins with the ideal generated by the entries of $C_{\text{comm}}(w)$:

$$I_{\text{comm}} = \langle (C_{\text{comm}}(w))_{ab} \rangle.$$

Part II (Section 4) gives the first algebraic data at the canonical point: $\dim T_1 \Sigma_{\text{comm}} = 7$, $\text{codim} = 11$, the kernel decomposition, and a computational tangent-local scaffold.

Remark 6.2 (Sparse Slices versus the Commutativity Locus).

Early two-dimensional scans found very few commutative sample points. Those experiments should be interpreted as transverse-slice evidence, not as a claim that Σ_{comm} is a sparse set in the full moduli space. The canonical point has a 7-dimensional computational tangent-local scaffold inside $\mathcal{M} = [0, 1]^{18}$. This distinction matters because the collision quotient and the accessibility jet are defined only on normal spectral charts inside Σ_{comm} , while the global A-spectrum remains defined even when a transverse path exits that locus.

Commutativity Locus

The collision quotient is defined only where the QT/HT averages commute

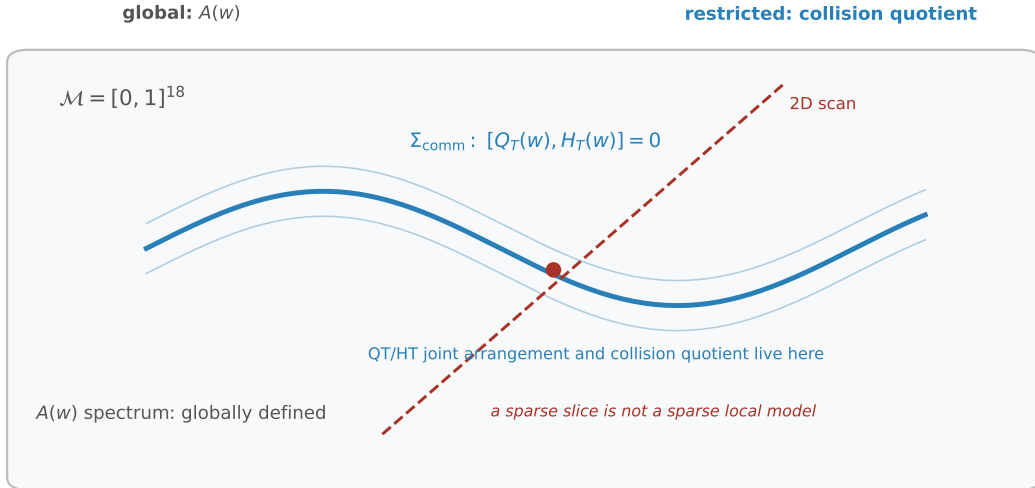


Figure 2. The commutativity locus is the algebraic domain controlling the QT/HT collision quotient. The global averaged spectrum is defined on the ambient weight space, while the joint arrangement and collision quotient are used only on normal commutative charts. Sparse transverse slices are evidence about crossings, not evidence that the commutative locus itself is sparse.

Definition 6.3 (Layer Wall).

Within the normal spectral domain, the layer wall is the locus where the collision quotient layer count is not locally constant:

$$\Sigma_L = \{w \in \Sigma_{\text{spec}} : N_L(w) \text{ jumps}\}.$$

Experimentally this is detected by **SPLIT** and **MERGE** events in $P(w)$.

Definition 6.4 (Field Wall).

Within the normal spectral domain, the field wall is the locus where the spectral field of the collision quotient changes:

$$\Sigma_{\text{field}} = \{w \in \Sigma_{\text{spec}} : \mathbb{Q}(L_\alpha(P(w))) \text{ changes}\}.$$

Experimentally this is detected when rational spectral data enter a mixed or $\mathbb{Q}(\sqrt{5})$ phase.

6.1 Conjecture 1 (Collision-Quotient Stratification)

For the collision-quotient observable, the expected internal hierarchy is

$$\Sigma_{\text{field}} \subseteq \Sigma_L \subseteq \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}.$$

Here Σ_{comm} is the primary commutator object, while Σ_L and Σ_{field} are spectral loci on normal charts inside it. The nesting is the organizing hierarchy for the algebraic theory. On the normal commutative spectral domain, field changes have been observed coincident with joint-spectrum bifurcations, supporting $\Sigma_{\text{field}} \subseteq \Sigma_L$.

This conjecture is specific to the collision quotient. The A-spectrum is a different global observable, and its field behavior can differ from the collision-quotient field behavior.

Part II – Local Geometry at the Canonical Point

7 Tangent Space of the Commutativity Locus

The commutativity locus is the zero set $C_{\text{comm}}(w) = 0$. At $w = \mathbf{1} = (1, \dots, 1)$, we have $C_{\text{comm}}(\mathbf{1}) = 0$ to machine precision ($\|[Q_T(\mathbf{1}), H_T(\mathbf{1})]\|_F = 3.77 \times 10^{-16}$). The first algebraic datum is the derivative $DC_{\text{comm}}|_{\mathbf{1}}$, giving the tangent space $T_{\mathbf{1}}\Sigma_{\text{comm}} = \ker DC_{\text{comm}}|_{\mathbf{1}}$.

7.1 Linearized Commutator Map

At $w = \mathbf{1}$:

$$Q_T(\mathbf{1}) = \frac{1}{12} \sum_{g \in \text{QT}} \rho(g), \quad H_T(\mathbf{1}) = \frac{1}{6} \sum_{g \in \text{HT}} \rho(g),$$

$$\text{with } [Q_T(\mathbf{1}), H_T(\mathbf{1})] = 0.$$

Lemma 7.1 (Jacobian at the canonical point).

For $g_k \in \text{QT}$:

$$J_k = \left. \frac{\partial C_{\text{comm}}}{\partial w_k} \right|_{\mathbf{1}} = [\rho(g_k), H_T(\mathbf{1})]/12.$$

For $g_k \in \text{HT}$:

$$J_k = \left. \frac{\partial C_{\text{comm}}}{\partial w_k} \right|_{\mathbf{1}} = [Q_T(\mathbf{1}), \rho(g_k)]/6.$$

Proof. For a weighted average $A(w) = \sum_i w_i M_i / \sum_i w_i$, the derivative at $w = \mathbf{1}$ is $\partial A / \partial w_k|_{\mathbf{1}} = (M_k - A(\mathbf{1})) / |S|$ where $|S|$ is the number of terms. Applying this to Q_T (12 terms) and H_T (6 terms) and using $[Q_T(\mathbf{1}), H_T(\mathbf{1})] = 0$ gives the stated formulas.

The tangent map sends $\delta w \in \mathbb{R}^{18}$ to the skew-Hermitian matrix $\sum_{k=1}^{18} \delta w_k J_k$. The tangent space is its kernel:

$$T_1 \Sigma_{\text{comm}} = \ker(\delta w \mapsto \sum_k \delta w_k J_k).$$

7.2 Tangent Space Dimension

Each J_k is a 228×228 skew-Hermitian matrix. A skew-Hermitian $N \times N$ matrix has $N(N-1)$ real degrees of freedom (the strictly upper triangular entries). For $N = 228$, this is $228 \cdot 227 = 51756$ real dimensions. Flattening each J_k to a real vector of length 51756 and assembling the 18 columns gives the 51756×18 real Jacobian matrix $\mathcal{J}$. Its singular value decomposition determines the rank and kernel, using the standard matrix-analysis interpretation of rank and singular values [15].

Theorem 7.2 (Computational Tangent Decomposition at the Canonical Point).

At $w = \mathbf{1}$, the tangent space $T_1 \Sigma_{\text{comm}} = \ker \mathcal{J}$ has the following properties.

(i) **Dimension.**

$$\dim T_1 \Sigma_{\text{comm}} = 7, \quad \text{codim} = 11.$$

(ii) **Singular value spectrum.** The 18 singular values of $\mathcal{J}$ form four nonzero groups and a zero group:

$$\begin{aligned} s_{0,1,2} &= 0.413880 & (\text{multiplicity } 3) \\ s_{3,4,5} &= 0.376796 & (\text{multiplicity } 3) \\ s_{6,7} &= 0.245327 & (\text{multiplicity } 2) \\ s_{8,9,10} &= 0.141639 & (\text{multiplicity } 3) \\ s_{11,\dots,17} &= 0 & (\text{multiplicity } 7). \end{aligned}$$

(iii) **Turn-class symmetry.** All 12 QT Jacobians have identical Frobenius norm $\|J_k\|_F = 0.309320$. All 6 HT Jacobians have identical norm 0.426730. This uniformity reflects the O_h (octahedral) conjugation symmetry of the full generator set.

(iv) **Kernel decomposition.** The 7-dimensional kernel splits by turn class:

$$\ker \mathcal{J} = K_{\text{HT}} \oplus K_{\text{QT}}, \quad \dim K_{\text{HT}} = 1, \quad \dim K_{\text{QT}} = 6.$$

K_{HT} is the uniform half-turn direction: all six HT weights scaled equally. K_{QT} is a 6-dimensional subspace of the 12-dimensional QT weight space. The subspace K_{QT} contains the three per-axis QT directions ($\{R, R', L, L'\}$ on axis 0, $\{U, U', D, D'\}$ on axis 1, $\{F, F', B, B'\}$ on axis 2) and their linear combinations (including uniform QT scaling).

(v) **Tested finite-amplitude tangency.** All seven kernel directions remain inside the commutative locus in the tested one-parameter families. Writing $w(\varepsilon) = \mathbf{1} + \varepsilon v$ for each unit kernel vector v , the full commutator norm satisfies

$$\|[Q_T(w(\varepsilon)), H_T(w(\varepsilon))]\|_F < 10^{-15}$$

for all tested $\varepsilon \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$. No kernel direction shows an observed ε^2 or higher-order departure within this tested range.

(vi) **First-order prediction.** For any weight direction v , the first-order approximation $\|\sum_k v_k J_k\| \cdot \varepsilon$ matches the true commutator norm to relative error $< 10^{-3}$ at $\varepsilon = 10^{-2}$ across all tested directions. At $\varepsilon = 0.1$, the ratio remains within 1% (actual/predicted $\in [0.990, 1.007]$ for the strongest transverse directions). The commutator map is nearly linear over a substantial neighborhood of $\mathbf{1}$.

(vii) **Local rank stability.** The Jacobian rank was recomputed at points $w = \mathbf{1} + \varepsilon v$ for random linear combinations v of the seven kernel basis vectors, at $\varepsilon \in \{0.01, 0.1, 0.5, 1.0\}$. In all tested cases the rank remains exactly 11 – the same as at $\mathbf{1}$. No rank increase is observed anywhere on the tested kernel family. This gives no evidence for hidden constraints that would increase the codimension of Σ_{comm} near $\mathbf{1}$ within the tested local model.

(viii) **Absence of nonlinear kernel contributions.** A systematic search tested 200 random directions in $\mathbb{R}^{18}$ (including directions with substantial transverse component, $\text{trans_frac} > 0.5$), each on a grid of 500 ε -steps from 10^{-6} to 0.5. For every direction, the commutator norm minimum occurs at the smallest ε and scales linearly with the transverse component. Zero directions with non-negligible transverse component achieved $\|[Q_T, H_T]\|_F < 10^{-13}$. This search found no evidence of hidden nonlinear kernel elements – points on Σ_{comm} that are not tangent to the first-order kernel.

Together, (v)–(viii) establish a computational tangent-local scaffold for a 7-dimensional model of Σ_{comm} near the canonical point. The Jacobian has rank 11 at $\mathbf{1}$, the tested kernel families show rank stability, no nonlinear constraints are observed in the random searches, and the commutator map is approximately linear over the tested neighborhood. Thus the commutator ideal I_{comm} has 11 observed independent first-order constraints at $\mathbf{1}$, with no observed higher-order degeneracy in the tested families. A proof of algebraic smoothness or a complete set of defining equations for Σ_{comm} is not claimed here. The constant-rank viewpoint [19] and the local ideal language from commutative algebra and algebraic geometry [10, 14] provide the background for this scaffold.

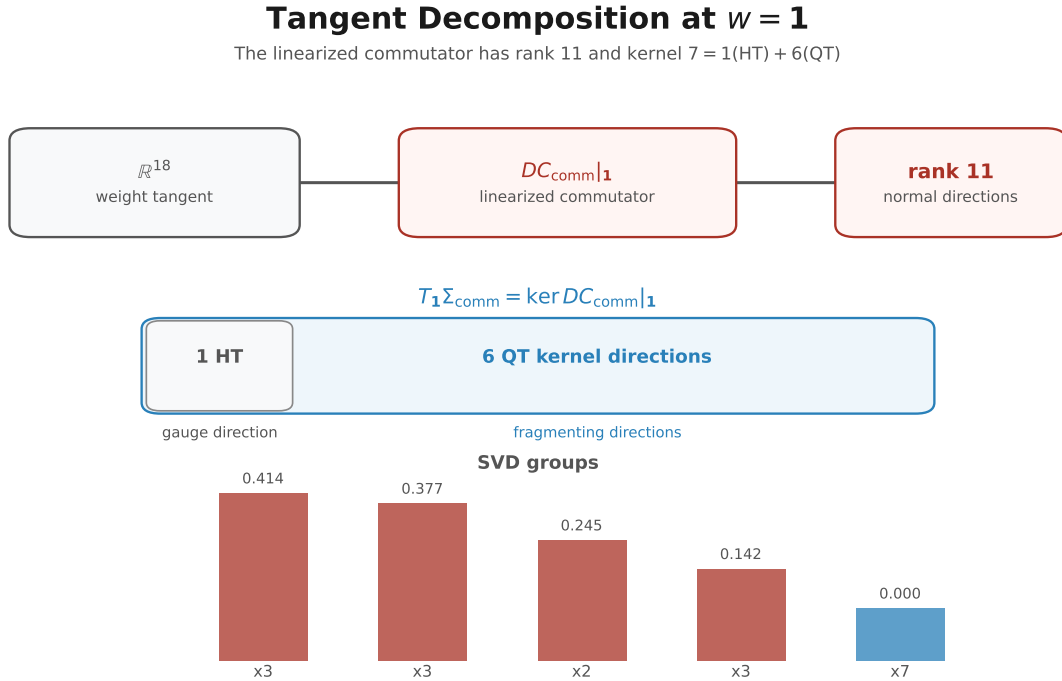


Figure 3. Computational tangent decomposition at the canonical point. The linearized commutator map has rank 11 and a 7-dimensional kernel, splitting in the tested local model into one uniform HT direction and six QT kernel directions.

7.3 Agreement with Numerical Slices

The tangent space classification perfectly predicts the commutator rank strata observed in transverse and symmetric one-parameter slices:

Direction class	In $T_1 \Sigma_{\text{comm}}$?	$\ [Q_T, H_T] \ _F$ at $\varepsilon = 1$
All QT symmetric	Yes	$< 10^{-16}$ (exact)
All HT symmetric	Yes	$< 10^{-16}$ (exact)
Per-axis QT symmetric ($\times 3$)	Yes	$< 10^{-16}$ (exact)
Per-axis HT symmetric ($\times 3$)	No (projection 0.577, residual 0.816)	~ 0.37 (rank 52)
Single QT deletion ($\times 12$)	No (projection 0.707, residual 0.707)	~ 0.31 (rank 96)
Single HT deletion ($\times 6$)	No (projection 0.408, residual 0.913)	~ 0.43 (rank 88)

7.4 Consequences for the Global Picture

The tangent space theorem corrects the interpretation of sparse two-dimensional slices. The (w_{F_2}, w_{B_2}) plane – both coordinates are axis-0 HT weights – is completely transverse to Σ_{comm} . The intersection $\Sigma_{\text{comm}} \cap \{2\text{D plane}\}$ is the single point $w = \mathbf{1}$. The “extreme sparsity” (1/961 cells) is an artifact of transverse slicing: the 2D plane cuts through the moduli space in a direction transverse to the 7-dimensional Σ_{comm} .

Conversely, the verified symmetry slices contained in $K_{\text{QT}} \oplus K_{\text{HT}}$ (for example, varying all QT and HT weights uniformly) lie inside the tested normal spectral subfamilies of Σ_{comm} and have a well-defined collision quotient at every tested point. A global algebraic description of this local family would require deriving the defining equations of I_{comm} beyond the tangent model.

7.5 Fragmentation Theorem: The Wall Origin Principle

Theorem 7.2 supplies a computational 7-dimensional tangent-local scaffold for Σ_{comm} at $\mathbf{1}$. A natural next question is whether the sector decomposition itself, namely the joint eigenspace projectors $P_i(w)$ of $(Q_T(w), H_T(w))$, is stable along the kernel directions. The answer is sharply negative, and this negation contains the mechanism behind all subsequent accessibility walls.

Theorem 7.3 (Computational Fragmentation at the Canonical Point).

At $w = \mathbf{1}$, the joint spectral decomposition has 9 sectors with dimensions $(20, 2, 39, 66, 26, 39, 1, 27, 8)$. The following statement is computational and local to the seven tested SVD kernel-basis directions:

(i) Normalized HT gauge versus SVD kernel direction. *The true gauge is uniform HT scaling after the QT/HT operators are normalized: it leaves $H_T(w)$ and $Q_T(w)$ invariant, so the 9-sector decomposition is preserved. The SVD basis vector labelled `kernel[0]` is tangent to this HT-scaling direction at $\mathbf{1}$, but it is not itself the normalized gauge curve at finite amplitude. Consequently it preserves the 9 sectors at $\varepsilon = 10^{-6}$ but fragments at larger ε in the table below.*

(ii) Immediate fragmentation. *All six QT kernel-basis directions (1–6) fragment the 9-sector decomposition at all tested small ε values. The fragmentation is hierarchical:*

Kernel direction	Sector count at $\varepsilon = 10^{-6}$	Sector count at $\varepsilon = 0.1$
kernel[0] (HT-tangent basis direction)	9 (tested gauge tangent)	35
kernel[1] (axis-2 QT prime)	24	35
kernel[2] (axis-0 QT prime)	25	35
kernel[3] (axis-1 QT prime)	31	35
kernel[4] (QT mixed)	32	35
kernel[5] (QT mixed)	34	35
kernel[6] (QT mixed)	24	35

Each kernel direction produces a distinct intermediate sector count at small ε , revealing a fine stratification of fragmentation types within Σ_{comm} . At larger ε (≥ 0.1), most directions converge to 35 sectors – the generic count in the tested kernel families.

(iii) Block-rank jumps. The fragmentation directly causes accessibility wall crossings. At $\varepsilon = 0.01$ along a generic kernel direction, the sector count jumps $9 \rightarrow 35$. In the directed support graph E_1^{dir} , the edge count jumps from 29 to 391; in the generator-labelled directed graph E_1^{gen} , the edge count jumps from 438 to 6334. After the initial fragmentation, the block-rank structure is stable until the next wall crossing.

Wall Origin Principle. Accessibility walls are not caused by nonlinearity of Σ_{comm} . They are caused by combinatorial instability of the sector decomposition under linear motion of the joint spectral points $(q_i(w), h_i(w))$. The mechanism is:

1. Σ_{comm} has a computational 7-dimensional tangent-local model at $\mathbf{1}$ (Theorem 7.2).
2. On normal spectral charts $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$, the joint spectral points move linearly with w (to first order, verified to $< 1\%$ at $\varepsilon = 0.01$).
3. The sector decomposition – the assignment of joint eigenspaces to distinct (q, h) points – undergoes discrete combinatorial changes when spectral points cross, merge, or split.
4. These discrete changes in the sector decomposition cause discrete changes in R_1 , R_2 , and D .

Thus the accessibility phase diagram on Σ_{spec} is a **piecewise-constant structure on a smooth linear background**. The walls are the loci where the joint spectral point arrangement undergoes a combinatorial transition.

(iv) The canonical point as a fragmentation intersection. The point $w = \mathbf{1}$ is special: it sits at the intersection of fragmentation walls for all six QT kernel directions and along finite-amplitude continuations of the HT-tangent SVD basis direction. The 9-sector coarse decomposition exists only at the canonical point in the tested QT kernel families, except along the exact normalized HT gauge curve. Every tested small QT kernel perturbation resolves the 9 sectors into finer sectors. The canonical Rubik spectrum is therefore a **maximally coarse point** on the normal spectral chart inside Σ_{comm} – the point where the largest number of joint spectral points have coincident $L_{2/3}$ values, producing the 6-layer collision quotient.

Part III – Accessibility Theorem Layer

The preceding sections describe the spectral geometry of the generator-set moduli space. We now fix the objects needed for the accessibility theorem layer. Throughout this part, w lies in a normal chart of $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$, so the QT/HT joint spectral arrangement and its orthogonal sector projectors are defined.

Fragmentation and Wall Origin

A computational smooth local model can carry discontinuous sector and accessibility shadows

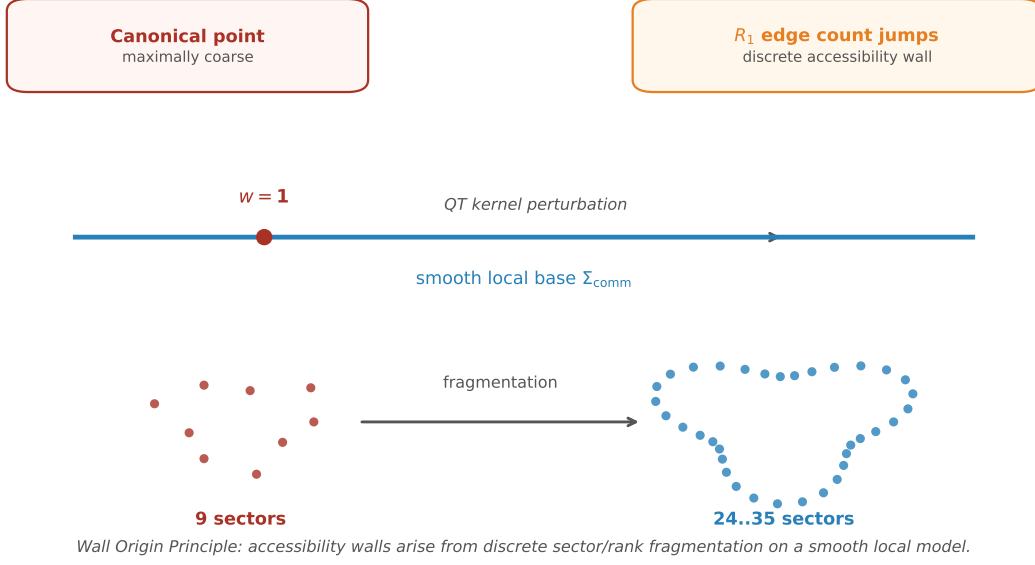


Figure 4. Fragmentation and the Wall Origin Principle. Along tested kernel directions, the commutativity base has a tested tangent-local model while the sector decomposition and projected accessibility data jump discretely.

8 Sectorized Systems and Accessibility Data

Definition 8.1 (Sectorized System over the Commutativity Locus).

Let $w \in \Sigma_{\text{spec}}$. The commuting normal pair $(Q_T(w), H_T(w))$ has an orthogonal joint spectral decomposition

$$V = \bigoplus_i V_i(w), \quad Q_i(w) : V \rightarrow V_i(w),$$

where $Q_i(w)$ denotes the orthogonal projector onto the i th joint sector. Let

$$X_g = \log \rho(g)$$

denote the skew-Hermitian accessibility generator used in the Paper V Lie-depth calculus. The sectorized system at w is the data

$$\mathcal{S}(w) = (V, \{Q_i(w)\}_i, \{X_g\}_{g \in S}).$$

This definition is only made on normal spectral charts. Outside Σ_{comm} , the QT/HT joint sectors are not defined; inside Σ_{comm} but outside the normal spectral domain, the present paper does not use orthogonal projector-valued sectorized accessibility data.

Definition 8.2 (Generator Block Maps).

For $w \in \Sigma_{\text{spec}}$, sectors i, j , and generator $g \in S$, define the sector block map

$$B_{ij}^g(w) = Q_i(w) X_g Q_j(w).$$

These blocks are the local matrix entries of the Lie accessibility generator with respect to the moving joint-sector decomposition. They are the Paper VI moving-sector analogue of the Paper V block maps.

Definition 8.3 (R-One Support Graph).

The first accessibility observable is the colored generator-support graph $R_1(w)$. Its vertices are the joint sectors i . For each generator $g \in S$, there is a colored directed edge

$$i \xrightarrow{g} j$$

if and only if

$$B_{ij}^g(w) = Q_i(w)X_gQ_j(w) \neq 0.$$

Thus $R_1(w)$ records the sector pairs directly touched by individual generators. It is a support-level object; it does not record cancellations in commutators or higher words.

We use three related edge-count conventions:

$$E_1^{\text{gen}}(w) = \{(g, i, j) : B_{ij}^g(w) \neq 0\},$$

the **generator-labelled directed graph**. Forgetting the generator label gives the **directed support graph**

$$E_1^{\text{dir}}(w) = \{(i, j) : \exists g, B_{ij}^g(w) \neq 0\}.$$

For display only, forgetting orientation gives the simple **support graph**

$$E_1^{\text{supp}}(w) = \{\{i, j\} : \exists g, B_{ij}^g(w) \neq 0\}.$$

The observable $R_1(w)$ itself is the generator-labelled directed graph $E_1^{\text{gen}}(w)$. Whenever a numerical edge count is quoted, the convention is stated explicitly.

Definition 8.4 (R-Two Repair Graph).

The second accessibility observable is the projected commutator-survival graph $R_2(w)$. For generators $g, h \in S$, define

$$C_{ij}^{g,h}(w) = Q_i(w)[X_g, X_h]Q_j(w).$$

The R_2 graph records the labeled pair (g, h) from sector j to sector i when

$$C_{ij}^{g,h}(w) \neq 0.$$

Equivalently, $R_2(w)$ records which length-two commutator candidates survive after projection to the moving joint-sector decomposition. This is the first repair layer beyond the support graph $R_1(w)$.

Definition 8.5 (Minimal Accessibility Depth).

The minimal accessibility depth data $D(w)$ records, for each ordered sector pair (i, j) , the least depth at which the pair becomes accessible under the chosen accessibility calculus. In the present theorem layer, depth 0 means direct generator support (R_1), depth 1 means first commutator survival (R_2), and larger depths belong to the higher Lie/PBW/word towers discussed in the accessibility framework.

We write

$$D(w) = (D_{ij}(w))_{i,j}.$$

The precise depth convention must be kept fixed within any theorem statement. For Paper VI, the role of $D(w)$ is to detect whether the moving sectorized system changes the gap between Lie-generated accessibility and composition-generated accessibility.

9 Accessibility Fields and Jets

The objects $R_1(w)$, $R_2(w)$, and $D(w)$ are discrete-valued functions on a tested local base. Their discontinuities are therefore not explained by ordinary differentiability of the functions themselves. The geometric object that mediates between the tested local commutativity model and the discrete accessibility data is the collection of matrix fields from which those discrete data are read.

Definition 9.1 (Accessibility Fields).

On a neighborhood $U \subset \Sigma_{\text{spec}}$ where the joint-sector projectors $Q_i(w)$ vary smoothly, define the accessibility fields:

$$B_{ij}^g(w) = Q_i(w)X_gQ_j(w),$$

$$C_{ij}^{g,h}(w) = Q_i(w)[X_g, X_h]Q_j(w),$$

and, for each Hall monomial M used in the finite-depth accessibility calculus,

$$H_{ij}^M(w) = Q_i(w)MQ_j(w).$$

The Hall monomial language is the finite-depth Lie algebra background inherited from Paper V [3], using Hall bases and free-Lie-algebra conventions [13, 21]. The Lie-generated accessibility viewpoint is aligned with the standard control-theoretic use of Lie brackets [18]; the basic Lie identities are those of the standard finite-dimensional Lie algebra setting [16].

The discrete observables are the support, commutator-survival, and first-depth shadows of these fields:

$$R_1(w) = \text{supp}(B(w)), \quad R_2(w) = \text{supp}(C(w)), \quad D(w) = \text{firstdepth}(H(w)).$$

Thus R_1, R_2, D are not fixed invariants attached once and for all to the representation. They are functions on normal spectral charts $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$, obtained by applying rank/support thresholds to smooth matrix fields.

It is convenient to bundle the first-order responses of these fields into the accessibility jet components

$$\mathcal{J}_{\text{acc}}(w) = (J_{\text{block}}(w), J_{\text{comm}}(w), J_{\text{depth}}(w)),$$

where

$$J_{\text{block}} = j^1 B, \quad J_{\text{comm}} = j^1 C, \quad J_{\text{depth}} = j^1 H.$$

Thus J_{block} is the generator-support response, J_{comm} is the Lie-defect tensor, and J_{depth} is the composition-propagation kernel.

Definition 9.2 (Accessibility Jet Bundle).

The accessibility jet at $w \in \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$ is the first-order data

$$\mathcal{J}_{\text{acc}}(w) = j_w^1(B, C, H),$$

where B, C, H denote the block, projected-commutator, and Hall projection fields above. For a tangent vector $v \in T_w \Sigma_{\text{comm}}$ represented by a local curve in Σ_{spec} , the accessibility jet gives the directional first variations

$$\nabla_v B_{ij}^g, \quad \nabla_v C_{ij}^{g,h}, \quad \nabla_v H_{ij}^M.$$

The discrete data R_1, R_2, D are shadows of these three jet components. They are locally constant precisely where the relevant matrix fields keep the same support and rank profile. Accessibility walls are therefore rank/support discriminants of the accessibility jet bundle. This terminology is intentionally geometric: it uses only first-order jet data of the underlying matrix fields, in the standard sense of jet-bundle geometry [22].

Theorem 9.3 (Jet-to-Accessibility Reduction).

Let $w_0 \in \Sigma_{\text{spec}}$ be a point where the commutator map has stable Jacobian rank, no hidden kernel emergence, first-order linearization accuracy, and a normal spectral chart in a neighborhood of w_0 . Let $v \in T_{w_0} \Sigma_{\text{comm}}$ be represented by a local curve $w(\varepsilon) \subset \Sigma_{\text{spec}}$ with $w(0) = w_0$ and $w'(0) = v$. Then the jet bundle $\mathcal{J}_{\text{acc}}$ is well defined in that neighborhood, and the QT/HT pair has a first-order expansion

$$(Q_T(w(\varepsilon)), H_T(w(\varepsilon))) = (Q_T(w_0), H_T(w_0)) + \varepsilon J(v) + O(\varepsilon^2),$$

and the accessibility fields have first-order expansions

$$B(w(\varepsilon)) = B(w_0) + \varepsilon \nabla_v B + O(\varepsilon^2),$$

$$C(w(\varepsilon)) = C(w_0) + \varepsilon \nabla_v C + O(\varepsilon^2),$$

$$H(w(\varepsilon)) = H(w_0) + \varepsilon \nabla_v H + O(\varepsilon^2).$$

Consequently, the accessibility jet $\mathcal{J}_{\text{acc}}(w_0)$ gives the first-order continuation of the matrix data from which R_1, R_2, D are read.

Moreover,

$$R_1 = \Pi_1(\mathcal{J}_{\text{acc}}), \quad R_2 = \Pi_2(\mathcal{J}_{\text{acc}}), \quad D = \Pi_D(\mathcal{J}_{\text{acc}})$$

for the support, commutator-survival, and first-depth projection maps Π_1, Π_2, Π_D . These projection maps are measurable and discrete-valued: they forget the continuous matrix entries and retain only the rank/support data relevant to accessibility.

Interpretation. *The discrete observables themselves do not have linear Taylor expansions. Instead, their defining matrix fields do. On any region where these fields preserve rank and support, the discrete shadow (R_1, R_2, D) is locally constant. An accessibility wall is the locus where this first-order continuation fails to preserve the rank/support profile.*

10 Spectral Walls and Accessibility Walls

The spectral wall hierarchy is

$$\Sigma_{\text{field}} \subseteq \Sigma_L \subseteq \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}.$$

Here Σ_{comm} is the commutator wall, while Σ_{spec} is the normal commutative spectral domain on which the collision quotient and orthogonal sector projectors are used. Σ_L records layer-count jumps, and Σ_{field} records field changes. These are spectral stratification walls. They describe how the joint arrangement $P(w)$ moves, splits, merges, and changes arithmetic field.

They are not accessibility walls by definition.

Definition 10.1 (Accessibility Wall).

An accessibility wall is a locus inside a normal spectral chart $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$ such that

$$w \in \Sigma_{\text{access}} \iff \text{the rank/support profile of } \mathcal{J}_{\text{acc}}(w) \text{ is not locally constant.}$$

Equivalently, at least one of the discrete accessibility observables is not locally constant. We write

$$\Sigma_{\text{access}} = \{w \in \Sigma_{\text{spec}} : (R_1, R_2, D) \text{ is not locally constant}\}.$$

Equivalently, Σ_{access} is the union of the rank/support discriminants for the block, projected-commutator, and Hall projection fields. The three visible components are

$$\Sigma_{R_1} = \{w \in \Sigma_{\text{spec}} : R_1(w) \text{ is not locally constant}\},$$

$$\Sigma_{R_2} = \{w \in \Sigma_{\text{spec}} : R_2(w) \text{ is not locally constant}\},$$

and

$$\Sigma_D = \{w \in \Sigma_{\text{spec}} : D(w) \text{ is not locally constant}\}.$$

At the support level,

$$\Sigma_{\text{access}} = \Sigma_{R_1} \cup \Sigma_{R_2} \cup \Sigma_D.$$

The cumulative and residual wall convention used below refines this union by separating walls that occur after lower-level data have been held fixed. The main accessibility-theoretic problem is to understand the relationship between this accessibility discriminant and the spectral walls Σ_L and Σ_{field} inside Σ_{spec} .

Remark 10.2 (Spectral Walls Are Not Accessibility Walls).

A field change in the collision quotient is a spectral event. It may indicate that the arithmetic form of the moving arrangement has changed, and it may be correlated with accessibility changes. But this correlation is not a definition. The accessibility wall is detected only when $R_1(w)$, $R_2(w)$, or $D(w)$ changes.

This separation is essential for Paper VI. It prevents the theorem layer from identifying an arithmetic spectral transition with a Lie/composition accessibility transition before the corresponding $R_1/R_2/D$ data have been computed.

11 Accessibility Stratification

The central structural claim of Paper VI is not merely that Σ_{comm} admits a computational tangent-local scaffold (Theorem 7.2) or that sectors fragment (Theorem 7.3). It is that the accessibility data R_1, R_2, D form a **discrete stratification** of this local model, and that this stratification is combinatorial in origin – governed by block-rank jumps, not by nonlinearity of the local base.

11.1 Local Constancy Lemma (Computational)

Let $w_0 \in \Sigma_{\text{spec}}$ and let v be a tangent direction in $T_{w_0} \Sigma_{\text{comm}}$ represented by a local curve $w(\varepsilon) \subset \Sigma_{\text{spec}}$ with $w(0) = w_0$ and $w'(0) = v$. In the tested kernel families below, this curve is given by the verified finite-amplitude continuation used in the support scripts. A generic straight line $w_0 + \varepsilon v$ is not assumed to remain inside the commutativity locus.

Lemma 11.1 (Sector Projector Smoothness).

For sufficiently small ε , the joint-spectral projectors $Q_i(w(\varepsilon))$ vary smoothly (to first order, linearly) with ε . The joint spectral points $(q_i(\varepsilon), h_i(\varepsilon))$ satisfy $\|(q_i(\varepsilon), h_i(\varepsilon)) - (q_i(0), h_i(0))\| \propto \varepsilon$.

This is verified numerically: along kernel direction 1 at $w = \mathbf{1}$, the (q, h) -point drift satisfies $\text{drift}/\varepsilon \approx 0.07$ for all 9 points up to the fragmentation threshold ($\varepsilon \approx 3 \times 10^{-7}$).

Lemma 11.2 (Block-Rank Piecewise Constancy).

For a fixed sector pair (i, j) and generator g , the block map $B_{ij}^g(w) = Q_i(w)X_gQ_j(w)$ has a rank that is piecewise constant in ε . Rank jumps occur at discrete values of ε where the joint spectral geometry undergoes a combinatorial transition (sector count change, eigenvalue crossing, or subspace alignment).

This is verified numerically: along a generic kernel direction, the total generator-labelled edge count $|E_1^{\text{gen}}|$ jumps from 438 to 5346 at the first fragmentation threshold, then remains piecewise constant with additional discrete jumps at larger ε .

11.2 Accessibility Stratification Theorem

Theorem 11.3 (Accessibility Stratification on a Normal Commutative Chart).

Let $w_0 \in \Sigma_{\text{spec}}$ be a point where the joint-spectral sector projectors $Q_i(w)$ depend smoothly on w in a neighborhood $U \subset \Sigma_{\text{spec}}$. Then on U :

(i) **R_1 wall.** $R_1(w)$ is locally constant on the open set where all block ranks $\text{rank}(B_{ij}^g(w))$ are constant. A change in $R_1(w)$ occurs only when at least one block rank changes – that is, at the locus

$$\Sigma_{R_1} = \{w \in \Sigma_{\text{spec}} : \exists i, j, g : \text{rank}(B_{ij}^g(w)) \text{ jumps}\}.$$

(ii) **R_2 wall.** $R_2(w)$ is not determined by $R_1(w)$. Both are different discrete projections of the accessibility jet $\mathcal{J}_{\text{acc}}$: R_1 is read from the block-field support, while R_2 is read from projected commutator survival. With $R_1(w)$ fixed, the projected commutator blocks $Q_i(w)[X_g, X_h]Q_j(w)$ are computed from the accessibility fields, specifically the block fields and their projected commutators. $R_2(w)$ is locally constant on the open subset of the R_1 -stratum where all projected commutator block ranks are constant. A change in $R_2(w)$ while $R_1(w)$ is fixed occurs at the locus

$$\Sigma_{R_2}^\circ = \{w \in \Sigma_{\text{spec}} \setminus \Sigma_{R_1} : \exists i, j, g, h : \text{rank}(Q_i(w)[X_g, X_h]Q_j(w)) \text{ jumps}\}.$$

(iii) **D wall.** With $R_1(w)$ and $R_2(w)$ fixed, the Lie-depth matrix $D(w)$ is determined by the weighted Hall coefficients (Paper V). $D(w)$ is locally constant on the open subset of the (R_1, R_2) -stratum where all relevant Hall monomial projections have constant rank. A change in $D(w)$ while R_1, R_2 are fixed occurs at

$$\Sigma_D^\circ = \{w \in \Sigma_{\text{spec}} \setminus (\Sigma_{R_1} \cup \Sigma_{R_2}^\circ) : D(w) \text{ jumps}\}.$$

(iv) **Hierarchy.** The residual walls $\Sigma_{R_2}^\circ$ and Σ_D° record changes that occur after lower-level data are held fixed. The corresponding cumulative wall sets are

$$\widehat{\Sigma}_{R_1} = \Sigma_{R_1}, \quad \widehat{\Sigma}_{R_2} = \Sigma_{R_1} \cup \Sigma_{R_2}^\circ, \quad \widehat{\Sigma}_D = \Sigma_{R_1} \cup \Sigma_{R_2}^\circ \cup \Sigma_D^\circ.$$

Thus the accessibility observables have the cumulative hierarchy

$$\boxed{\widehat{\Sigma}_{R_1} \subseteq \widehat{\Sigma}_{R_2} \subseteq \widehat{\Sigma}_D = \Sigma_{\text{access}} \subseteq \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}.}$$

On the local domain $U \subset \Sigma_{\text{spec}}$, outside $\widehat{\Sigma}_D \cap U$, the full accessibility triple (R_1, R_2, D) is locally constant. Thus U carries a local stratification by the level sets of (R_1, R_2, D) , and these strata are separated by the cumulative wall loci. This is a local stratification on a normal chart of the commutativity locus; extending it to the ambient moduli space $\mathcal{M} = [0, 1]^{18}$ would require a separate theory.

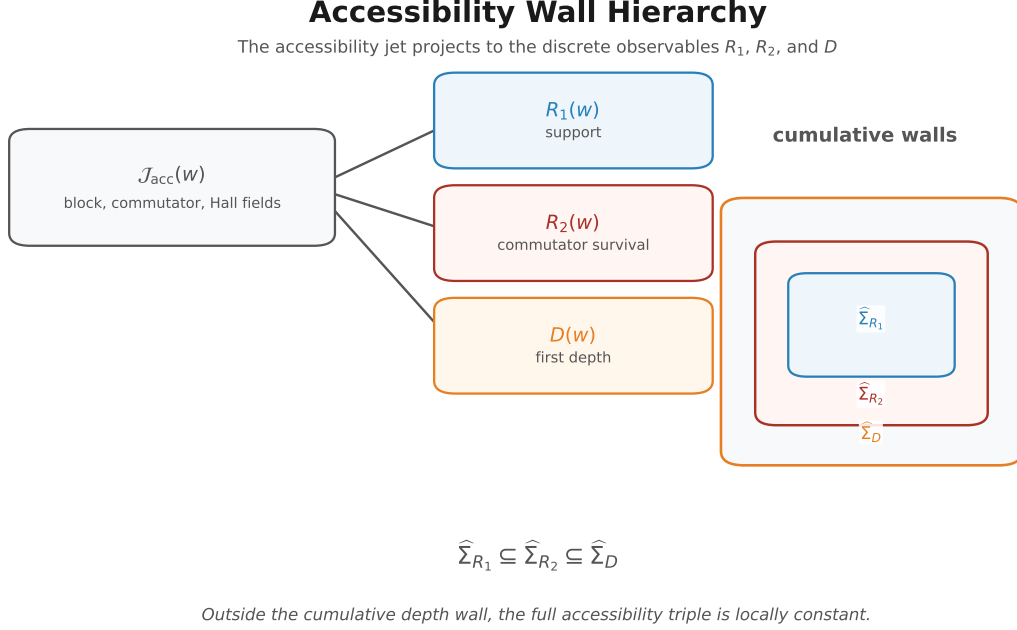


Figure 5. Accessibility wall hierarchy. The accessibility jet projects to the discrete observables R_1 , R_2 , and D ; the cumulative wall sets record where these rank/support shadows fail to continue locally.

(v) Wall origin. The residual and cumulative accessibility walls are not caused by nonlinearity of Σ_{comm} . They are caused by combinatorial instability of block projections under smooth (to first order, linear) deformation of the joint spectral points on Σ_{spec} . The mechanism is:

smooth motion of $(q_i, h_i) \rightarrow$ discrete jump in block rank $\rightarrow$ discrete change in $R_1/R_2/D$.

This is the **Accessibility Wall Mechanism**: accessibility walls describe where the projection of spectral geometry onto the sector decomposition ceases to be structurally invariant.

11.3 Lie-versus-Composition Phase Transition

The accessibility wall formalism is designed to detect changes in the gap between Lie-generated and composition-generated accessibility. Away from Σ_{access} , the local accessibility jet has constant rank/support profile, so the triple (R_1, R_2, D) is locally constant. In that region the comparison between Lie-generated accessibility and compositional accessibility is stable under small generator-set perturbations.

Crossing Σ_{access} changes this comparison. A jump in R_1 changes direct generator support. A residual jump in R_2 changes which length-two commutator candidates survive after projection while direct support is held fixed. A jump in D changes the minimal depth at which a sector pair becomes accessible. Thus accessibility walls are phase-transition loci for the Lie-versus-composition gap.

11.4 Computational Support

The stratification pattern is supported by three probes:

Rubik cube (18 generators, moving sectors). Along a generic kernel direction in the tested normal spectral chart:

ε	Sectors	$ E_1^{\text{gen}} $	Event
0 (canonical)	9	438	maximally coarse point
$\sim 10^{-7}$	24	4520	sector fragmentation
$\sim 10^{-5}$	24	4520	stable
$\sim 10^{-4}$	34	6264	secondary fragmentation
10^{-2}	35	6334	tertiary wall
0.1–1.0	35	6334	piecewise constant

Five distinct walls detected. The canonical point is a maximally coarse configuration: every tested small QT kernel perturbation triggers sector fragmentation.

S4-3gen-B (3 generators, fixed sectors). Varying the weight of generator c from 0 to 1:

Weight c	$ E_1^{\text{gen}} $	Event
0	15	degenerate (c-blocks vanish)
10^{-8}	32	first R_1 jump
10^{-6}	33	second R_1 jump
10^{-4} –1.0	33	piecewise constant

A5-3gen (3 generators, 15 sectors). Identical pattern:

Path	$ E_1^{\text{gen}} $ at $t = 0$	$ E_1^{\text{gen}} $ at $t > 0$	Jump at
c-weight scan	48	86	$t = 10^{-8}$
a-weight scan	76	86	$t = 10^{-8}$
uniform scan	0	86	$t = 10^{-8}$

Common pattern in tested systems. Across all three systems, regardless of mechanism (sector fragmentation in Rubik vs. block-norm crossing in S4/A5), the same structural pattern is observed: degenerate weight configurations at isolated or low-dimensional points in the tested scans reduce R_1 ; small perturbations trigger discrete R_1 jump(s); after the jump, R_1 is piecewise constant; walls form a nested hierarchy. This tested pattern suggests a general wall phenomenon – a combinatorial bifurcation on a continuous background, not a Rubik-specific accident.

12 Computational Reproducibility

The computational claims in this paper are supported by the following repository scripts, listed by role:

Label	Role
Moduli scan	global generator-set scans, collision quotient versus A-spectrum, and stabilized bifurcation tables
Tangent map	tangent model for $T_1\Sigma_{\text{comm}}$ and tested nonlinear-kernel searches
Fragmentation audit	sector fragmentation and R_1 wall formation along kernel directions
Wall-crossing summary	compact mechanism-level accessibility wall-crossing support table

The corresponding scripts live in `experiments/paper6/`, with a shared utility layer for phase tables. Exploratory searches and verbose demonstrations are retained in the repository for provenance, but the manuscript relies only on the summarized tables above.

13 Generator-Set Probes

The following tables are not separate theoretical inputs. They are finite probes of the wall picture developed above. Single-generator deletions test transverse directions in the ambient weight space: they show how easily one leaves Σ_{comm} , so the collision quotient becomes undefined even though the global A -spectrum continues to vary. The selected generator families test special lower-dimensional faces and slices: some remain inside the normal commutative spectral domain, while others cross the commutativity wall. Together they display the two-observable distinction that motivates Paper VI:

$$\begin{aligned} \text{collision quotient} &: \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}, \\ A(w)\text{-spectrum} &: \text{ambient weight space.} \end{aligned}$$

These probes also locate the spectral part of the wall hierarchy before the accessibility jet is applied. Once a path lies in a normal spectral chart Σ_{spec} , the moving sector projectors give accessibility fields and the question becomes whether the associated jet crosses Σ_{access} .

13.1 Single-Generator Deletions

All 18 single-generator deletions exit Σ_{comm} . The A -spectrum shows 14 layers (HT deletion) or 21 layers (QT deletion), all with mixed field. This is now understood via Theorem 7.2: single-coordinate directions have residual 0.707 (QT) or 0.913 (HT) outside the kernel.

13.2 Selected Generator Families

Family	CQ status	CQ layers	CQ field	A-layers	A-field
$n = 18$ full	defined	6	$\mathbb{Q}$	6	$\mathbb{Q}$
$n = 16$ drop axis-0 HT	blocked	–	undefined	9	mixed
$n = 15$ drop negative-face HT	blocked	–	undefined	23	mixed
$n = 14$ drop axis-1 QT	defined	10	mixed	8	mixed
$n = 12$ QT only	defined	6	$\mathbb{Q}$	6	$\mathbb{Q}$
$n = 6$ HT only	defined	3	$\mathbb{Q}$	3	$\mathbb{Q}$

14 Outlook

The present paper establishes the geometric framework and the computational local model at the canonical point. Several algebraic questions remain open. First, the global ideal I_{comm} should be reduced to explicit defining equations for the commutativity wall. Second, the rank strata of the commutator map should be characterized algebraically, rather than only through numerical signatures such as the observed ranks 0, 52, 88, 96. Third, the accessibility walls associated with R_2 and D should be computed on selected normal spectral strata $\Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$.

The cancellation and incidence mechanism examples inherited from Paper V show that R_1 can remain fixed while R_2 or D changes under perturbation. These examples motivate, but do not yet constitute, a general classification theorem for accessibility walls. Such a theorem would

require combining the commutator ideal of Σ_{comm} with the weighted Hall path data controlling finite-depth accessibility.

Paper VII [6] continues this same structure in the static direction. It treats the accessibility jet as the candidate completion object and identifies Type IV incidence as a high-codimension algebraic degeneration. Thus the three papers should be read as different projections of one framework: Paper V classifies the local mechanisms, Paper VI studies how their observable shadows move on normal spectral charts inside Σ_{comm} , and Paper VII isolates the generic completion boundary.

15 Conclusion

Paper IV studies collision quotients of a fixed joint arrangement. Paper VI shows what changes when the generator set itself varies: the arrangement becomes $P = P(w)$, the collision quotient is defined only on the normal commutative spectral domain, and the accessibility data carried by the moving sectors become piecewise-constant observables with their own walls.

At the canonical Rubik point, the commutativity wall has a computationally verified 7-dimensional tangent model with codimension 11. Along the tested kernel-family spectral charts, the canonical nine-sector decomposition is maximally coarse: tested QT kernel directions fragment the sectors and trigger discrete changes in R_1 . Spectral walls describe how the arrangement moves; accessibility walls describe when that motion changes the gap between Lie-generated and composition-generated accessibility.

This gives a common language for the later papers. A point of Σ_{spec} determines a sectorized operator system

$$\mathcal{S}(w) = (V, \{Q_i(w)\}, \{X_g\}_{g \in S}),$$

and $R_1(w)$, $R_2(w)$, $D(w)$, and $\mathcal{J}_{\text{acc}}(w)$ are derived fields of this block-operator geometry. Paper V studies these fields for a fixed sectorized system. Paper VI studies their variation over the commutative deformation base. Paper VII [6] studies generic completion away from the incidence strata where the block geometry becomes nongeneric.

Bibliographic Notes

Program lineage. Paper VI depends internally on Papers I–V and the CCS: Papers I–III define the canonical Rubik spectral, sector, transport, and accessibility data [8, 7, 4, 9]; Paper IV supplies the fixed-arrangement collision-quotient viewpoint [5]; Paper V supplies the local $R_1/R_2/D$ repair calculus and weighted Hall path language [3]. The reproducibility scripts are listed explicitly in the Computational Support section.

External background. Finite-group representations and the Rubik group setting use the standard references [23, 17]. The fixed/moving joint-spectrum side is closest to finite-dimensional spectral algebra and algebraic combinatorics, with association-scheme and Bose–Mesner algebras serving as comparison classes rather than asserted identifications [2, 11]. Quotients of association schemes and modern Bose–Mesner computational tools provide neighboring formalisms for quotient structures in commutative semisimple matrix algebras [12, 20]; here they are cited as background, not as an identification of the QT/HT algebra with an association scheme. The local geometry of Σ_{comm} uses smooth-manifold, jet-bundle, and local algebraic-geometry language [19, 22, 14, 10]. The accessibility layer uses Hall bases, free Lie algebras, standard Lie algebra identities, and geometric-control accessibility language [13, 21, 16, 18, 1]. Matrix-rank and rank-stability arguments are interpreted in the standard matrix-analysis setting [15].

Computational provenance. The Paper VI support suite consists of the moduli scan, tangent map, fragmentation audit, wall-crossing summary, and shared phase-table utilities. Exploratory searches and verbose demonstrations remain in the repository for provenance, but the manuscript relies only on those summarized tables.

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